

Syed Ahsan Masud Zaidi

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Research Interests

Computer Vision · Video Understanding · Machine Learning · Deep Learning · Vision Transformers · Athlete Safety & Injury Prevention · Evolutionary Computing · Signal Processing

Education

Doctor of Philosophy — Computer Science

2023 – Expected 2027

Kansas State University, Manhattan, KS, USA

PhD Candidate — CGPA: 4.0 / 4.0

- Advisor: Prof. Lior Shamir (collaborators: Prof. William H. Hsu, Dr. Scott Dietrich, Albright College)
- Research: Computer Vision & Video Understanding; Zero-Shot Event Localization; Action Classification with Vision Transformers; Athlete Safety & Injury Prevention

Master of Science — Electrical Engineering

2017 – 2019

King Fahd Univ. of Petroleum & Minerals (KFUPM), Dhahran, Saudi Arabia

CGPA: 3.75 / 4.0 — QS #67

- Thesis: *Precise Position Control of Laser Beam for Free Space Optics Communications*
- Focus: Control Systems, Artificial Intelligence
- Awarded competitive full merit scholarship

Bachelor of Science — Mechatronics & Control Engineering

2009 – 2013

University of Engineering & Technology (UET), Lahore, Pakistan

CGPA: 3.565 / 4.0

- Thesis: *Static Marker-Based Cricket Bowling Analysis*
- Focus: Machine Vision, Embedded Systems, Robotics

Publications

Peer-Reviewed Conference Papers

3. **Zaidi, S.A.M.**, Shamir, L., Hsu, W.H., Dietrich, S., & Zaidi, T. (2026). *GRAZE: Grounded Refinement and Motion-Aware Zero-Shot Event Localization*. **CVSports Workshop @ CVPR 2026** (Spotlight Presentation). [arXiv:2604.01383](#) · [Code: GitHub](#).
2. **Zaidi, S.A.M.**, Hsu, W.H., & Dietrich, S. (2026). *ViTs for Action Classification in Videos: An Approach to Risky Tackle Detection in American Football Practice Videos*. **International Conference on Pattern Recognition (ICPR 2026)**; also presented at **CVSports Workshop @ CVPR 2026** (Poster). [arXiv:2604.01318](#) · [Code: GitHub](#).
1. Khan, R.A., Asad, A., Jamal, M.S., & **Zaidi, S.A.** (2018). *Olympus Field Development Optimization Study — KFUPM*. **EAGE/TNO Workshop on OLYMPUS Field Development Optimization**, cp-562-00019.

Preprints / Under Review

1. Nayab, S., Chohan, S.R., Jameel, A., Shah, S.R., **Zaidi, S.A.M.**, Jha, A.N., et al. (2025). *Advancing Remote and Continuous Cardiovascular Patient Monitoring Through a Novel and Resource-Efficient IoT-Driven Framework*. [arXiv:2505.03409](#). (Under Review)

Thesis

Zaidi, S.A.M. (2019). *Position Control of Laser Beam for Free Space Optics Communication*. M.Sc. Thesis, KFUPM. PQDT-Global.

Academic & Professional Experience

PhD Candidate / Graduate Research & Teaching Assistant

2023 – Present

Department of Computer Science, Kansas State University, Manhattan, KS

- Developed GRAZE, a zero-shot video event localization framework; spotlight presentation at CVSports @ CVPR 2026
- Developed ViT-based risky tackle detection system for American football; accepted at ICPR 2026 and CVSports @ CVPR 2026
- Instructor of Record, CIS-209: C/Python Programming for Engineers (Spring 2026) — sole instructor responsible for course design, lectures, assessments, and student evaluation
- GTA, CIS-209: C/Python Programming for Engineers (2023–Fall 2025) — led lab sessions, held office hours

Lecturer

2014 – 2023

Department of Mechatronics & Control Engineering, UET Lahore, Pakistan

- Designed and delivered 10+ undergraduate courses: Embedded Systems, Digital Logic Design, Signals & Systems, Electric Circuits, Electronic Devices & Circuits, Introduction to Computer Programming, Engineering Dynamics, Manufacturing Processes, Fundamentals of Thermal Sciences
- Director, Simulation Lab; In-Charge, Semester Cell

Teaching Assistant / Lab Instructor

2017 – 2019

Department of Electrical Engineering, KFUPM, Dhahran, Saudi Arabia

- EE-555: Neural Networks Theory & Applications
- EE-663: Image Processing
- PHYS-101 & PHYS-102: Engineering Physics Laboratory

Project Engineer

2013 – 2014

RBL PepsiCo, Lahore, Pakistan

- Commissioned a new KRONES PET filling plant; managed system integration and production handover

Founder & CEO

2014 (6 months)

Black Gold Solutions, Plan 9 — Punjab Information Technology Board, Lahore, Pakistan

- Founded technology startup at Pakistan's largest tech incubator (Plan 9, PITB)
- Third Prize, TIE Startup Challenge 2014

Selected Research Projects

- **GRAZE** (2025–26): Zero-shot video event localization via grounded refinement and motion-aware temporal reasoning; uses Grounding DINO and SAM2 for pixel-level contact verification; spotlight at CVSports @ CVPR 2026
- **Risky Tackle Detection** (2025–26): ViT-based (ViViT) action classification for dangerous tackle detection in American football; Taguchi-guided augmentation; accepted at ICPR 2026 and CVSports @ CVPR 2026
- **Seismic Event Detection**: Deep CNN with face portrait representations for intelligent passive seismic event classification from time-series sensor data

- **PV Peak Power Prediction:** Genetic Algorithm-based MPPT optimizer for photovoltaic panels using evolutionary computation
- **Free Space Optics Control** (M.Sc. Thesis): Laser beam position control system incorporating computer-vision-based feedback for FSO communication links
- **OLYMPUS Reservoir Optimization:** Field development optimization using reservoir simulation; published at EAGE/TNO Workshop 2018
- **Cricket Bowling Analysis** (B.Sc. Thesis): Static marker-based biomechanical analysis of cricket bowling technique using machine vision
- **Autonomous Robotics:** Maze-solving, line-following, wall-chasing robots; FPGA-based beam balancing; 3× national first prizes

Technical Skills

Languages	Python, MATLAB, C, C++
ML / CV	PyTorch, TensorFlow, OpenCV, Scikit-learn, HuggingFace Transformers
Tools	LabVIEW, Git, Linux, L ^A T _E X
Hardware	Xilinx FPGA, Arduino, NI MyRIO, PCB Design
Specialisms	Vision Transformers (ViTs), Temporal Grounding, Zero-Shot Localization, Evolutionary Algorithms, Control Systems, Signal Processing

Honors, Awards & Grants

- **AOSSM Playmaker Grant Finalist**, American Orthopaedic Society for Sports Medicine (2026) — competitively selected to pitch AI-driven athlete safety research to the AOSSM Research Committee; Annual Meeting, Seattle, WA, July 2026
- **Broadening Participation in Computer Vision Grant**, Computer Vision Foundation / CVPR 2026 — competitive travel grant
- **Department Travel Award**, Department of Computer Science, Kansas State University (2026) — to present at ICPR 2026, Lyon, France
- **IAPR Travel Stipend & Registration Scholarship**, International Association for Pattern Recognition, ICPR 2026 — competitive award in recognition of accepted paper
- Full Merit Scholarship, M.Sc., KFUPM, Saudi Arabia (international competitive award)
- First Prize — All-Pakistan ROBOMAZE Competition, IEEE Society GIKI
- First Prize — ROBOTIC RACE, NAGS-11, GCU
- First Prize — Spades IEEE Robotic Race 2011, LUMS
- Third Prize — TIE Startup Challenge 2014
- Dell Laptop Award — Chief Minister Punjab, Pakistan (academic merit)
- 100% Merit Scholarship (Intermediate) — Punjab College of Sciences, Lahore
- Dr. Naseer Scholarship (Matriculation) — Unique Group of Institutions, Lahore

Presentations & Talks

- **Spotlight + Poster:** “GRAZE: Grounded Refinement and Motion-Aware Zero-Shot Event Localization,” CVSports Workshop @ CVPR 2026, Denver, CO, June 2026
- **Poster:** “ViTs for Action Classification in Videos: Risky Tackle Detection in American Football,” CVSports Workshop @ CVPR 2026, Denver, CO, June 2026
- **Oral Presentation:** “ViTs for Risky Tackle Detection in American Football,” K-State Graduate Research, Arts, and Discovery (KGRAD) Forum, Kansas State University, April 2026
- **Poster:** “AI for Athlete Safety and Injury Prevention,” K-State AI Symposium, Kansas State University, October 2025
- Department Seminar: LabVIEW for Engineering Applications, UET Lahore
- Department Seminar: NI MyRIO Embedded Platform, UET Lahore

- Department Seminar: PCB Manufacturing, UET Lahore

Professional Memberships & Service

Society Memberships

- Member, Computer Vision Foundation (CVF)
- Member, IEEE Technical Committee on Pattern Analysis and Machine Intelligence (TC-PAMI)
- Member, International Association for Pattern Recognition (IAPR)
- Member, Institute of Electrical and Electronics Engineers (IEEE)

Reviewing

- Reviewer, AI4Science Workshop @ ICML 2026
- Reviewer, *SLAS Technology* (journal)